

1 ASIO Full Analysis Report

This report summarizes the analysis of dataset 20250918.0x0002. The trial spans 599.49 s of continuous data acquisition across all six channels. The remainder of this section reports both raw and detrended + EMI-filtered statistics. Units are femtoamps for mean/RMS/std; skew and kurtosis are dimensionless.

1.1 Detrended + EMI-filtered Statistics

Table 1: Detrended + EMI-filtered Current Statistics

Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
SXR1	-1.9210×10^{-15}	4.9256×10^1	4.9256×10^1	1.9061×10^{-2}	3.8583×10^{-2}
SXR2	-2.3101×10^{-15}	7.0353×10^1	7.0353×10^1	6.5926×10^{-1}	2.2708×10^1
SXR3	-6.3934×10^{-16}	4.9370×10^1	4.9370×10^1	-6.5312×10^{-2}	8.1111×10^{-3}
SXR4	-1.2676×10^{-15}	6.5073×10^1	6.5073×10^1	-8.3927×10^{-3}	-1.8823×10^{-1}
HXR	-2.4804×10^{-15}	6.4431×10^2	6.4431×10^2	-1.3318×10^{-2}	7.2836×10^{-3}
EUV	-2.5552×10^{-14}	8.3542×10^2	8.3542×10^2	2.8915×10^{-2}	-8.6469×10^{-2}

1.2 Raw Statistics

Table 2: Raw Current Statistics

Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
SXR1	5.4265×10^4	5.4265×10^4	6.2986×10^1	8.3631×10^{-2}	-1.4157×10^{-1}
SXR2	5.3759×10^4	5.3760×10^4	8.3070×10^1	6.3527×10^{-1}	1.3882×10^1
SXR3	5.3992×10^4	5.3992×10^4	5.5859×10^1	-1.5888×10^{-2}	1.1705×10^{-1}
SXR4	5.3238×10^4	5.3238×10^4	7.1618×10^1	5.2573×10^{-4}	-2.3516×10^{-1}

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Channel	Mean (fA)	RMS (fA)	Std Dev (fA)	Skew	Kurtosis
HXR	6.7941×10^5	6.7941×10^5	7.4329×10^2	-3.9512×10^{-2}	3.4925×10^{-2}
EUV	7.6807×10^5	7.6807×10^5	1.0985×10^3	2.0833×10^{-1}	-2.3160×10^{-1}

1.3 Trial Duration

Quantity	Value
Total acquisition time	599.490 s

1.4 Raw Voltage Time Series

ASIO Raw Voltage Values

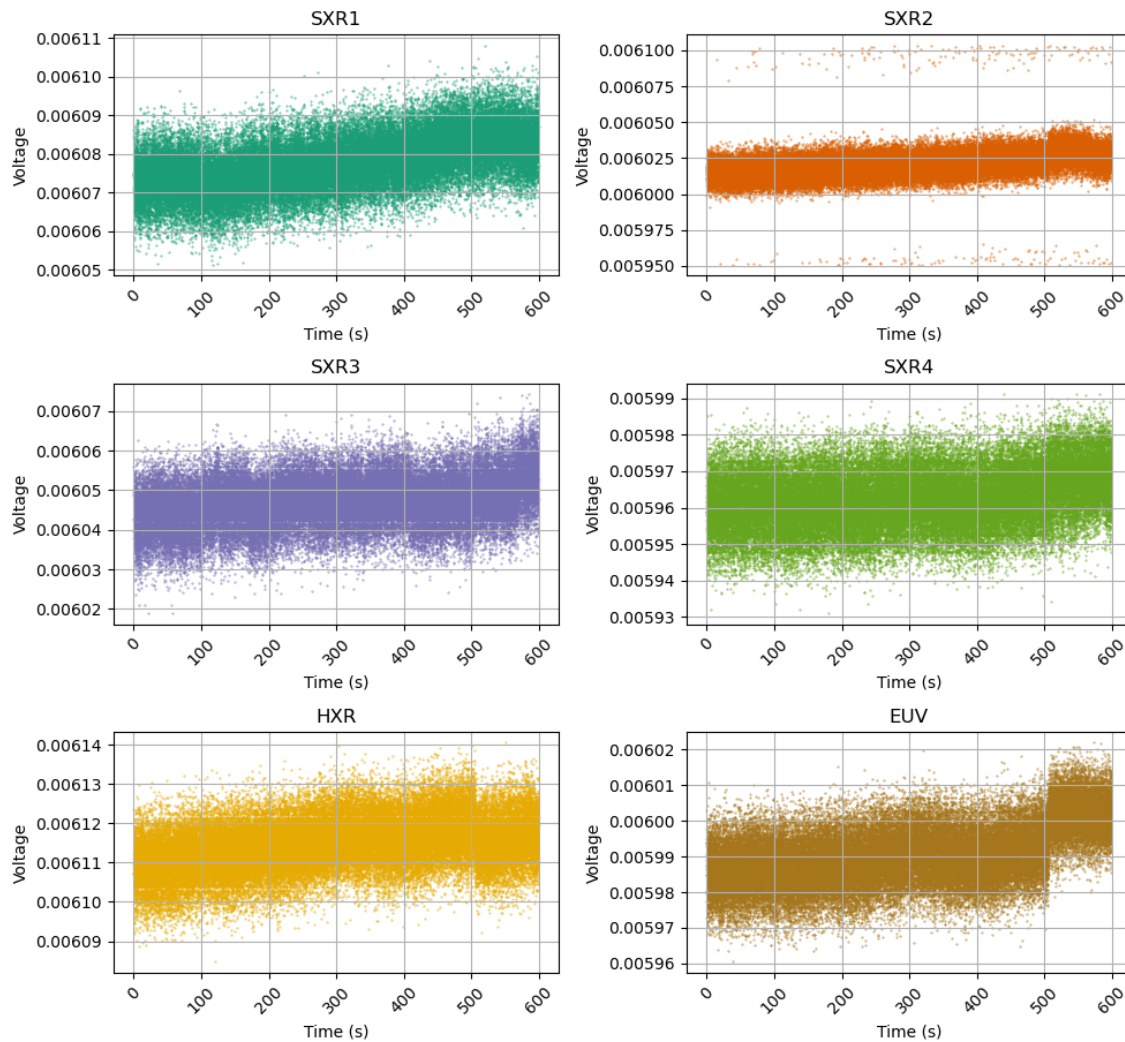


Figure 1: Raw voltage vs. time (s) for all six ASIO channels.

1.5 FFTs of Raw Voltages

FFTs

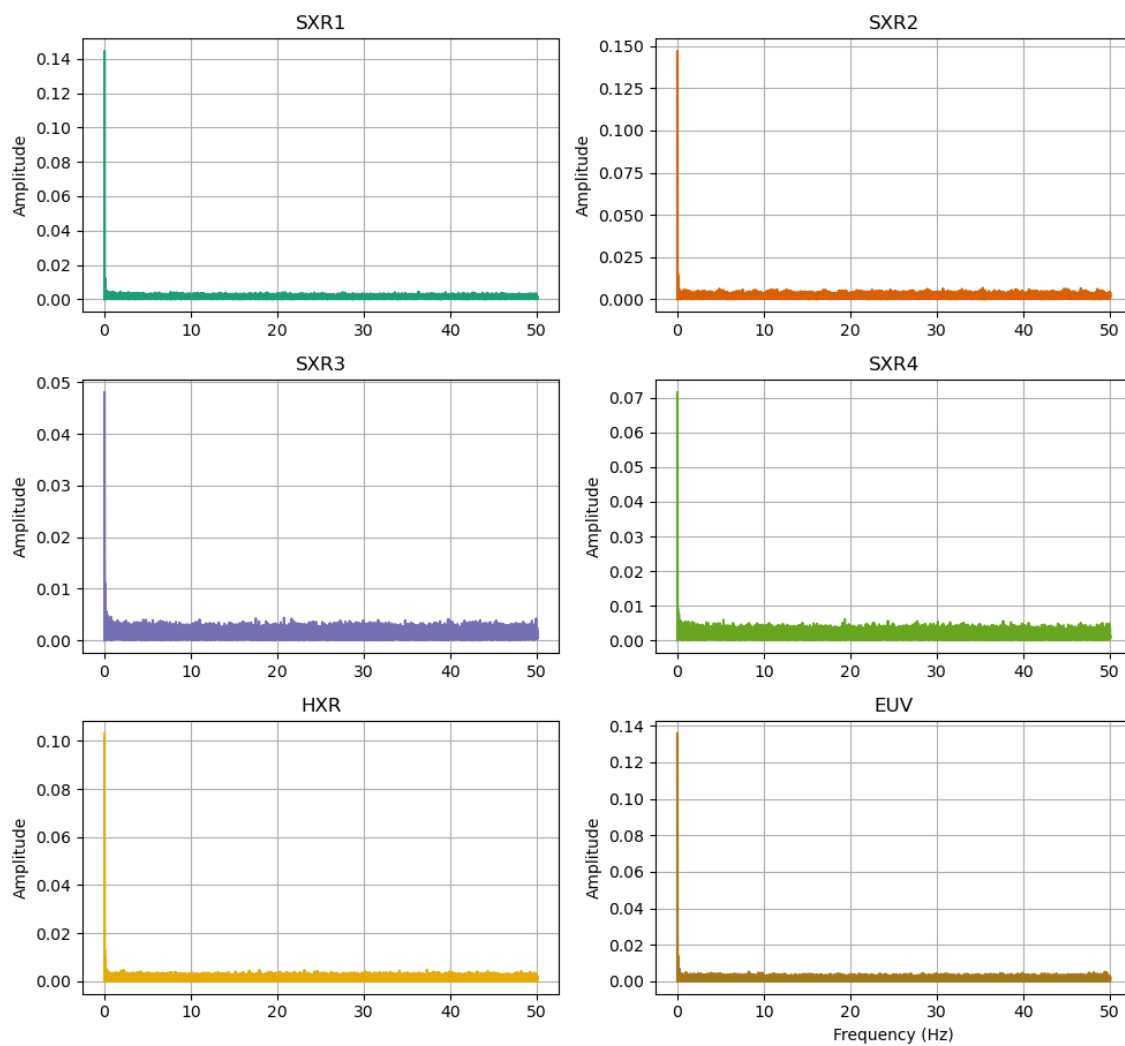


Figure 2: FFTs of raw voltages (mean centered).

1.6 Detrended + EMI-filtered Time Series

EMI-Filtered and Detrended Voltage Values

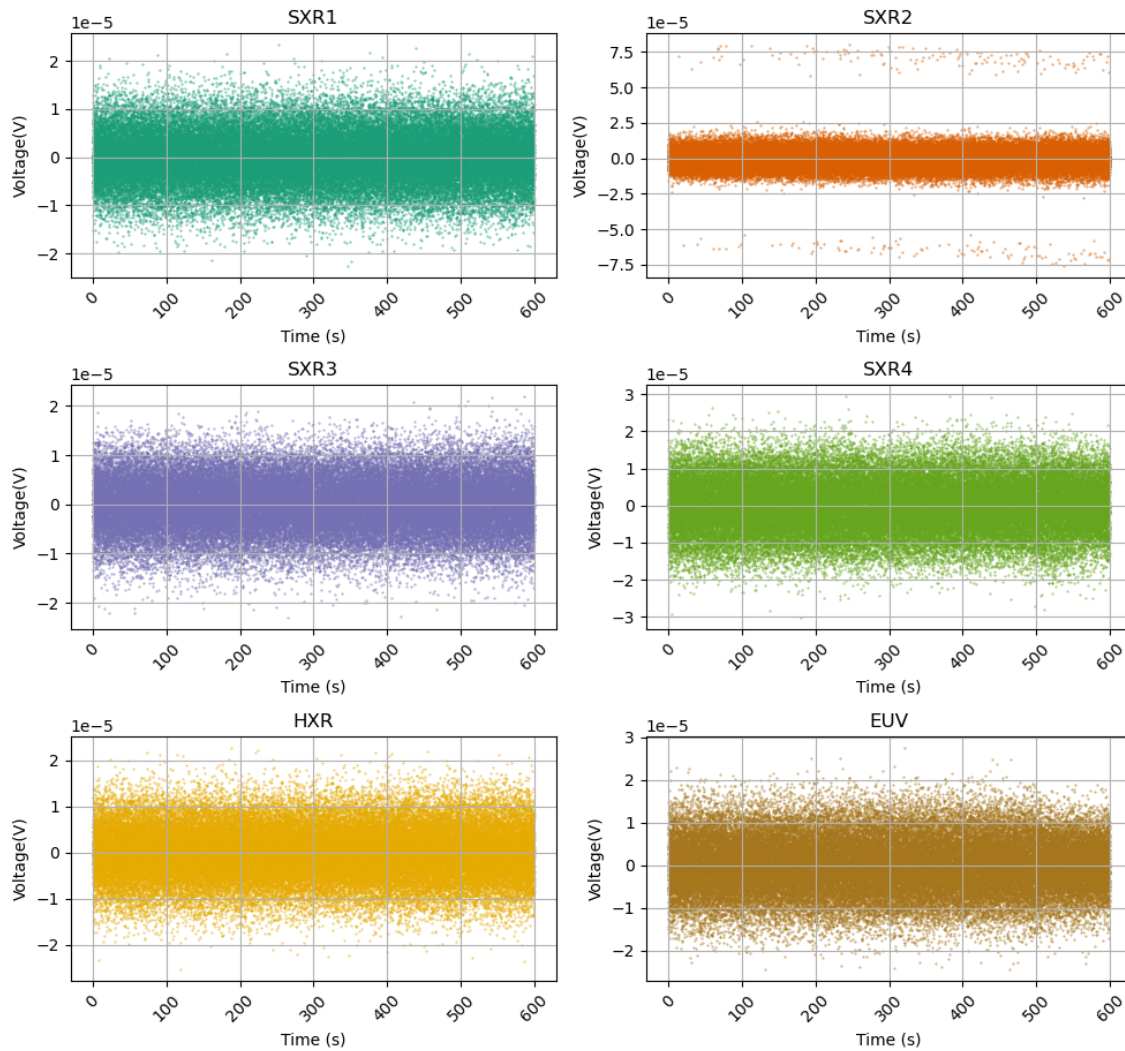


Figure 3: Voltage vs. time (s) after Savitzky–Golay detrending and 40 Hz EMI removal.

1.7 Noise Histograms

ASIO Noise Histograms w/EMI-filtering and Detrending

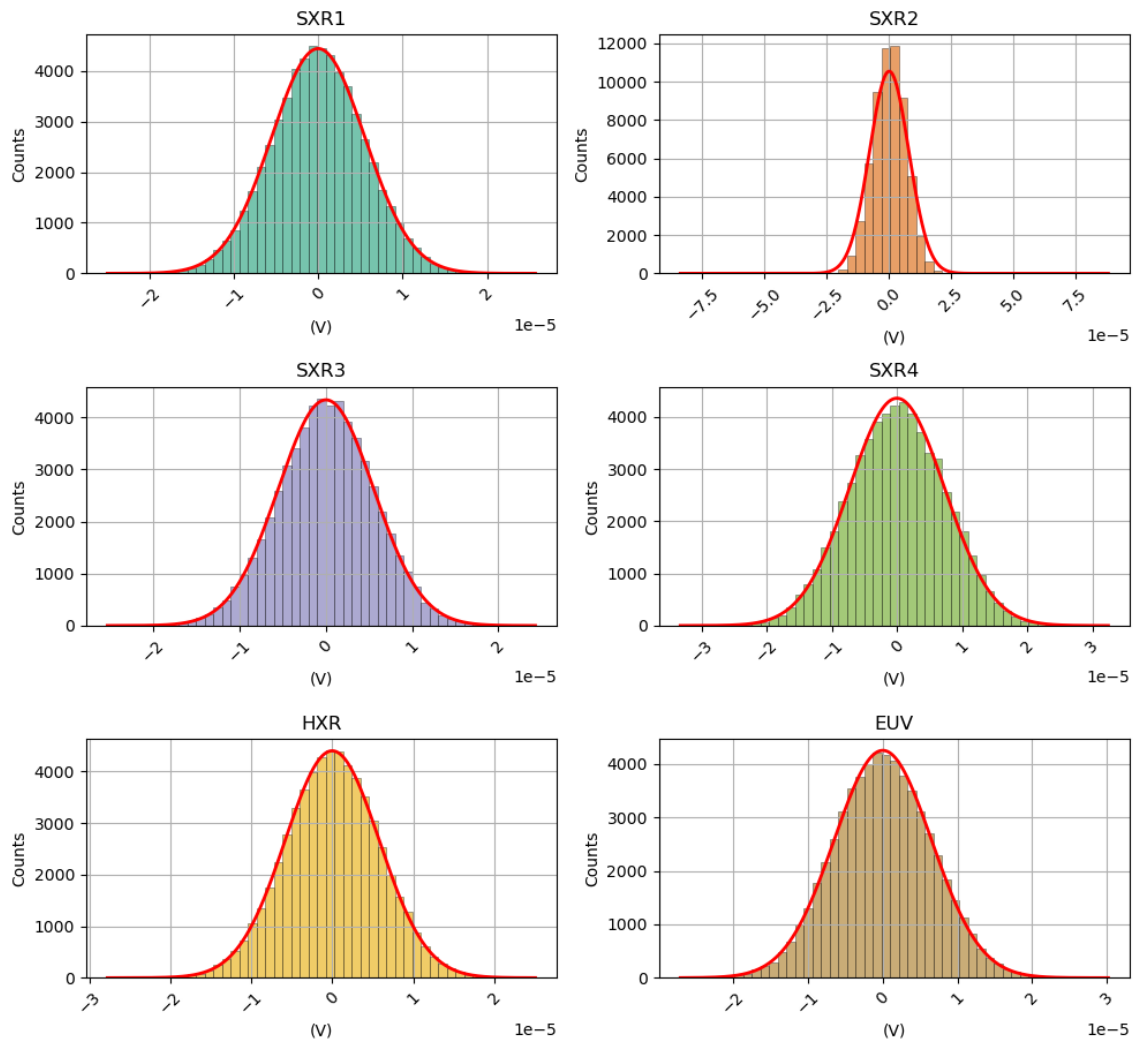


Figure 4: Noise histograms after detrending and 40 Hz EMI removal. Red curve is a Gaussian fit.

1.8 Fe-55 Expectation Values

Fe-55 expectation values: not yet implemented.

Testing Notes

selection: whole dataset (t=0.0s–599.5s) Background test